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Algorithms for the Analysis of Biological Sequences 2025/2026

Sequence Metrics and Similarity

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Context and Motivation

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Applications Note
OXFORD

Phylogenetics

Nextstrain: real-time tracking of pathogen evolution

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Abstract

Summary: Understanding the spread and evolution of pathogens is important for effective public health measures and surveillance. Nextstrain consists of a database of viral genomes, a bioinformatics pipeline for phylodynamics analysis, and an interactive visualization platform. Together these present a real-time view into the evolution and spread of a range of viral pathogens of high public health importance. The visualization integrates sequence data with other data types such as geographic information, serology, or host species. Nextstrain compiles our current understanding into a single accessible location, open to health professionals, epidemiologists, virologists and the public alike.
Availability and implementation: All code (predominantly JavaScript and Python) is freely available from github.com/nextstrain and the web-application is available at nextstrain.org.
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1 Introduction

Viral pathogens pose an ever-present danger to global human health, highlighted by recent events such as the West African Ebola epidemic and the ongoing Zika epidemic in the Americas. The rapid evolution of these viruses allows inference of epidemic history from genomic data. Such analyses are often done in isolation, and may lack the spatial or temporal context in which to best interpret the results (Pybus *et al.*, 2013). Furthermore, the results of analyses are rarely made available to the public or health bodies until after publication, which may be too late to aid understanding or effect change in policy. We have developed Nextstrain to visualize outbreaks in as close to real time as possible. Whilst currently encompassing a selection of viruses, extension to non-viral pathogens is forthcoming.

The regularly updated nature and rapidity of these analyses is crucial to the monitoring and understanding of pathogen epidemiology and evolution. Sequencing times and costs are continually dropping, with on-the-ground sequencing used during recent epidemics (Faria *et al.*, 2017; Quick *et al.*, 2016). Rapid methods by which to analyze, interpret, and disseminate results must complement this speed of sequencing.

Nextstrain consists of data curation, analysis and visualization components: Python scripts maintain a database of available sequences and related metadata, sourced from public repositories such as NCBI (www.ncbi.nlm.nih.gov), GISAID (www.gisaid.org) and ViPR (www.viprbrc.org), as well as GitHub repositories and other sources of genomic data. A suite of tools perform phylodynamic analysis (Volz *et al.*, 2013), including subsampling, alignment, phylogenetic inference, temporal dating of ancestral nodes and discrete trait geographic reconstruction, including inference of the most likely transmission events. This leverages the maximum likelihood

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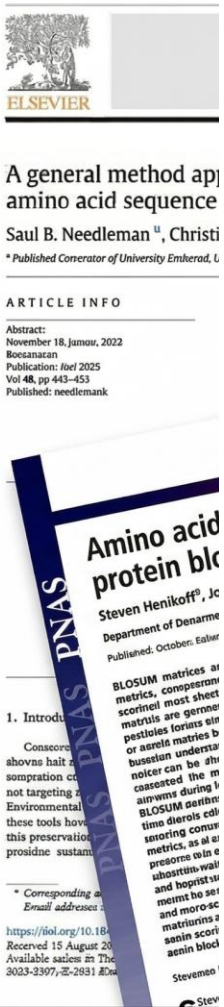

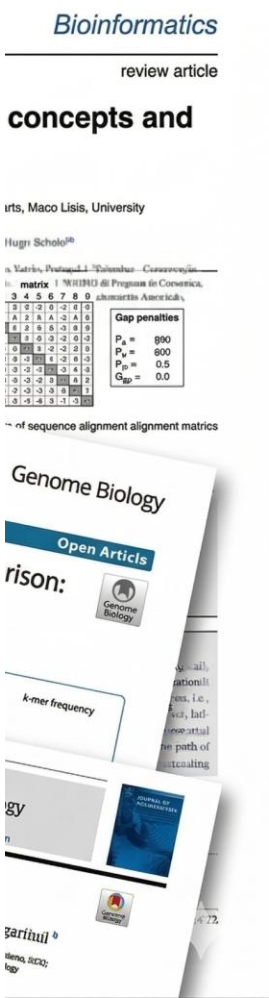

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What defines a metric

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Definition of a Metric

Metric – function which quantifies the distance between elements, satisfying 5 fundamentals axioms.

NON-NEGATIVITY

$d(x,y) \geq 0$

Distance is always non-negative.

IDENTITY OF INDISCERNIBLES

$d(x,y) = 0 \leftrightarrow x = y$

Zero distance means sequences are identical.

SYMMETRY

$d(x,y) = d(y,x)$

Distance is the same in both directions.

B
A C

TRIANGLE INEGUALITY

$d(x,z) \leq d(x,y) + d(y,z)$

The direct distance is always ≤ an indirect path.

B
A C

ULTRAMETIC INEQUALITY

$d(x,z) \leq \max(d(x,y), d(y,z))$

A stronger version of the triangle inequality.

Comparison Metrics

Raw Aligned Sequences (Raw Inputs with Gaps) – Example

Seq A:	G	A	G	G	T	A	C	G	G	C	G	T	G	A	T
Seq B:	G	T	G	G	T	A	A	C	G	G	G	G	T	A	T

Hamming Distance

Column-by-column Comparison

- Fixed-length only.
- Substitution only.
- Position-wise comparison.
- Not suited for the comparison of raw genomes.

Edit Distance

Operation-based Comparison

- Variable-length allowed.
- Insertion, deletion and substitution.
- Minimum-edit comparison.
- Well-suited for raw genome comparison.

In this study

- Hamming distance ≈ 20.000 (first 29.000 bp)
- Edit distance = 153 edits

Sequence Alignment

NC_045512.2 Severe acute respiratory syndrome coronavirus 2 isolate Wuhan-Hu-1, complete genome

Sequence ID: **Query_7430555** Length: **29903** Number of Matches: **1**

Range 1: 1 to 29903 Graphics ▼ Next Match ▲ Previous Match

NW Score	Identities	Gaps	Strand
59130	29750/29903(99%)	114/29903(0%)	Plus/Plus

```
Query  1     ---------------------------------------------------CTTGTAGATCT  10
                                                                ||||||||||
Sbjct  1     ATTAAAGGTTTATACCTTCCCAGGTAACAAACCAACCAACTTTCGATCTCTTGTAGATCT  60

Query  11    GTTCTCTAAACGAACTTTAAAATCTGTGTGGCTGTCACTCGGCTGCATGCTTAGTGCACT  70
             ||||||||||||||||||||||||||||||||||||||||||||||||||||||||||||
Sbjct  61    GTTCTCTAAACGAACTTTAAAATCTGTGTGGCTGTCACTCGGCTGCATGCTTAGTGCACT  120

Query  71    CACGCAGTATAATTAATAACTAATTACTGTCGTTGACAGGACACGAGTAACTCGTCTATC  130
             ||||||||||||||||||||||||||||||||||||||||||||||||||||||||||||
Sbjct  121   CACGCAGTATAATTAATAACTAATTACTGTCGTTGACAGGACACGAGTAACTCGTCTATC  180

Query  131   TTCTGCAGGCTGCTTACGGTTTCGTCCGTTTTGCAGCCGATCATCAGCACATCTAGGTTT  190
             |||||||||||||||||||||||||||||| |||||||||||||||||||||||||||||||
Sbjct  181   TTCTGCAGGCTGCTTACGGTTTCGTCCGTGTTGCAGCCGATCATCAGCACATCTAGGTTT  240

Query  191   TGTCCGGGTGTGACCGAAAGGTAAGATGGAGAGCCTTGTCCCTGGTTTCAACGAGAAAAC  250
              ||||||||||||||||||||||||||||||||||||||||||||||||||||||||||||
Sbjct  241   CGTCCGGGTGTGACCGAAAGGTAAGATGGAGAGCCTTGTCCCTGGTTTCAACGAGAAAAC  300

Query  251   ACACGTCCAACTCAGTTTGCCTGTTTTACAGGTTCGCGACGTGCTCGTACGTGGCTTTGG  310
             ||||||||||||||||||||||||||||||||||||||||||||||||||||||||||||
Sbjct  301   ACACGTCCAACTCAGTTTGCCTGTTTTACAGGTTCGCGACGTGCTCGTACGTGGCTTTGG  360

Query  311   AGACTCCGTGGAGGAGGTCTTATCAGAGGCACGTCAACATCTTAAAGATGGCACTTGTGG  370
             ||||||||||||||||||||||||||||||||||||||||||||||||||||||||||||
Sbjct  361   AGACTCCGTGGAGGAGGTCTTATCAGAGGCACGTCAACATCTTAAAGATGGCACTTGTGG  420

Query  371   CTTAGTAGAAGTTGAAAAAGGCGTTTTGCCTCAACTTGAACAGCCCTATGTGTTCATCAA  430
             ||||||||||||||||||||||||||||||||||||||||||||||||||||||||||||
Sbjct  421   CTTAGTAGAAGTTGAAAAAGGCGTTTTGCCTCAACTTGAACAGCCCTATGTGTTCATCAA  480

Query  431   ACGTTCGGATGCTCGAACTGCACCTCATGGTCATGTTATGGTTGAGCTGGTAGCAGAACT  490
             ||||||||||||||||||||||||||||||||||||||||||||||||||||||||||||
Sbjct  481   ACGTTCGGATGCTCGAACTGCACCTCATGGTCATGTTATGGTTGAGCTGGTAGCAGAACT  540

Query  491   CGAAGGCATTCAGTACGGTCGTAGTGGTGAGACACTTGGTGTCCTTGTCCCTCATGTGGG  550
             ||||||||||||||||||||||||||||||||||||||||||||||||||||||||||||
Sbjct  541   CGAAGGCATTCAGTACGGTCGTAGTGGTGAGACACTTGGTGTCCTTGTCCCTCATGTGGG  600
```

GLOBAL OFFSET

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If the genomes are aligned, what metrics should we use?

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Sliding Window Algorithm

Windowed Divergence Algorithm

Input: Aligned reference sequence (*ref*) and variant sequence (*var*), Window size (*w)*
Output: Dictionary with *pos*, Hamming (*ham*), and Edit (*edit*) distances

```
aln_len ← length of ref; ref_pos_map ← [ ]; coord ← 0; res ← {pos: [ ], ham: [ ], edit: [ ]}
FOR EACH character 'c' in ref DO:
    Append coord to ref_pos_map;
    IF 'c' ≠ '-' THEN coord ← coord +1
FOR start FROM 0 TO (aln_len – w) STEP w DO:
    chunk_ref, chunk_var ← segments of ref and var from start to start + w (with gaps)
    raw_ref, raw_var ← remove gaps ('-') from chunk_ref and chunk_var
    midpoint ← ref_pos_map[start + w/2]
    Append midpoint to res[pos]
    Append Hamming(chunk_ref, chunk_var) to res[ham]
    Append Edit(raw_ref, raw_var) to res[edit]
end FOR
RETURN res
```

Hands-On

- **NextStrain study**

https://www.researchgate.net/publication/325344695_Nextstrain_real-time_tracking_of_pathogen_evolution

- **GitHub repository**

https://github.com/amarcos20/projeto_investigacao_algoritmos

Includes all algorithms, documentation, and the sliding window pipeline.

Conclusions

From Data to Knowledge

Distance metrics provide the essential bridge to interpret the biological variability described in scientific literature.

Algorithmic Flexibility

There's no "perfect" metric; choosing between Hamming, Edit Distance, etc. depends on whether we are analyzing point mutations (SNPs) or complex evolutionary events (Indels).

The Power of Dynamic Programming

DP algorithms are fundamental to ensuring efficiency and accuracy when aligning real genomic sequences.

The Big Picture

These tools are the first step toward personalized medicine diagnostics and real-time tracking of viral variants.